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Task Description:	Flame spread over solid fuels in an opposed-flow environment has been investigated for over four decades for understanding the fundamental nature of hazardous fire spread. The appeal for this configuration stems from the fact that flame spread rate remains steady, even if the flame itself may grow in size. For practical fire safety issues, however, wind-assisted flame spread is more relevant. However, these two regimes have always been studied in isolation without much effort to establish a connection, even though the underlying mechanism of flame spread is the same in all regimes. Sitting between the two regimes are high-residence time flames, as found in a low-velocity or quiescent microgravity environment. Residence time is the time spent by an oxidizer in the combustion zone. Such flames, which are of interest on their own merit due to fire safety issues in spacecrafts, offer some unique characteristics because of the high residence time. Radiation becomes dominant and, based on previous space experiments and analysis, we contend that a vigorously spreading flame on Earth becomes self-extinguishing in a microgravity environment under certain conditions such as the fuel thickness being greater than a critical value. The proposed research uses a comprehensive approach-- a novel experimental set up and a theoretical framework based on scaling and numerical modeling-- to investigate flame spread driven by varying residence time, from blow-off extinction in an opposed-flow configuration through high residence time flame to blow-off extinction in a concurrent-flow configuration. At the heart of this proposal is a novel but simple experiment where the residence time of the oxidizer can be controlled and high residence time flames can be established for a long duration (compared to drop towers). As a proof of concept, we have constructed a flame tower at SDSU in which, after a sample is ignited, the sample holder, placed in an open moveable cart, can be traversed at any desired speed upward or downward, creating an external flow that can augment or mitigate the buoyancy-induced flow. Preliminary results show that we can control the residence time and create flames in different regimes, including a transition between a wind-aided and wind-opposed configuration. At Gifu University in Japan, we have been developing an interferometry based imaging system which we intend to enhance to capture the thermal footprint of a flame's leading edge. The leading edge is central to our understanding of mechanism of flame extinction. Further development of this technology will enable us to integrate diagnostics in future space based experiments and provide validation data to a comprehensive numerical model. The comprehensive model, to be built upon our existing two-dimensional model, will solve an unsteady, three-dimensional, Navier stokes equation with finite rate kinetics in the gas and solid phases and radiation in the gas phase. The software implementation will be object-oriented and utilize a new technology called Web Services that will decouple various sub-models and enhance parallel execution. The radiation model will also be refined by including the equilibrium composition of species for finding radiative properties in high residence-time flames. The new data generated by our proposed ground-based experiment will be the first of its kind to encompass flame spread across different regimes and will stimulate fundamental theoretical research to unify different regimes of flame spread. The comprehensive model, tested against available theory, data in literature, and data generated at SDSU and Gifu, will be applied to test the three hypotheses presented in this proposal regarding flame extinguishment in a microgravity environment. A successful outcome of this project will lead to a well thought out space-based experiment on the mechanism of flame extinction in a gravity free environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research has three components: We are building a novel experimental facility called Flame Tower at SDSU, where ground based experiments on flame spread will be conducted; Theoretical and computational work that explores the similarity and differences between the mechanisms flame spread in a zero gravity space environment and on Earth; Design and propose a space based experiment to establish extinction mechanism of flames. The data that we plan to acquire in the Flame Tower will provide the research community with a comprehensive set of results for testing different theories of flame spread in a normal gravity environment. Moreover, by controlling the residence time, various regimes of flame spread, including the microgravity regime, can be explored in the Flame Tower. Our theoretical work predicts a fuel thickness beyond which steady flame spread is unsustainable in a gravity free environment. If we are successful in establishing a critical thickness, this will have a powerful impact on making fire resistant environment for humans in space.
	The tasks we have performed to date can be separated into four different categories. Below, we list the progress we are making in each. A. Ground-Based Experimental Work: The goal of this work is to establish the role of residence time, time spent by an oxidizer in a flame leading edge, on the mechanism and control of flame spread. Towards this goal, we have been building a number of ground-based experiments involving flame spread over thin solid fuels in an opposed flow environment. A.1: SDSU Flame Tower: The flame tower is the centerpiece of our ground-based activities. We have finished the construction of a 10m tall steel chamber (details on year-1 report) inside which a fuel sample mounted on a cart can be traversed up or down a vertical rail with a prescribed velocity. We have been successful in developing a completely remote controlled system to move a cart at any command speed (from -3 m/s to + 3 m/s). Analysis of video images and data from velocity probes (details on year-2 report) confirm the success of our approach in creating a relative velocity with precision. A.2: MicroFlameTracker: The cart carrying the fuel sample must be self sufficient to ignite the sample, record flame spread rate in real time, acquire temperature and flow velocity data, take video of the flame from different angles - all these while being operated remotely from outside the chamber. This complex piece of equipment has been developed and successfully tested. An onboard microcontroller coupled with an onboard laptop (all operated by onboard battery) communicates with an outside command computer through a wireless network. The command computer also communicates wirelessly with the motion system that creates the desired cart velocity. Instead of describing the details of our design and fabrication, let us describe how the system is supposed to accomplish our goal. To begin an experiment, a hatch door is opened, and a new sample is mounted on the MicroFlameTracker. After the hatch is closed, the MicroFlameTracker is sent to one end of the tower, say, to the top for a downward run. Using Windows Remote Desktop, the command computer gains access of the onboard laptop. A Webcam (coupled with a LED light -the sealed tower is completely dark) focuses on a measuring tape attached to the tower guide rail, which ascertains the current location of the cart. Through a graphical command application developed by us, we can access the

Task Progress:	data recording system, ignition system, the flame tracking mechanism, and the video camera. We first turn on the data acquisition system, controlled by the onboard microcontroller, which also turns on a digital stopwatch. The video recorder, a high-resolution CCD camera, is then turned on which records the clock in the background as it captures the propagation of the flame from ignition to extinction. This allows us to synchronize the video data with all other data. The ignition command is sent and a Kanthal (iron-chromium-aluminum, FeCrAl) alloy igniter attached to the sample holder is heated by a circuit controlled by the microcontroller. The ignition wire is in contact with one end of the fuel sample which is secured in a magnetized frame with a rectangular cut out. Once ignition is visually confirmed from the Webcam window, the igniter is turned off. The tracking system consists of a thermocouple riding on a linear motion system parallel to the fuel sample. A PID control algorithm ensures that the thermocouple temperature remains unchanged by moving it in the direction of spread. As the thermocouple tracks the thermal front of the flame, the motion of the cart at a desired velocity is activated. For a velocity of, say, 20 cm/s, the cart takes about 35 seconds to traverse the effective length of the rail. As soon as the run is over, instantaneous spread rate (from the motion 0f the thermocouple) data is available. We have just finished our first set of preliminary experiments and by the next reporting period we expect to acquire a large set of data on flame spread rate vs. the flow velocity (both positive and negative). A.3: Flame Stabilizer: One of the challenges in the experimental study of flame spread is that, even if the flame spreads at a steady rate, the propagating flame creates an unsteady phenomenon with respect to the laboratory frame of reference. As a result, it is difficult to obtain detailed data, necessary for validating models, in a spreading flame. To remedy this situation, we have built a novel flame spread apparatus that moves the fuel in the opposite direction of the flame spread to keep the leading edge of the flame stationary with respect to the laboratory. A thermocouple, fixed to the laboratory frame of reference, in front of the leading edge of the flame senses the presence of the flame and a PID controller keeps its temperature constant by moving the sample holder, driven by a stepper motor, in the opposite direction at the velocity of the spread. Instantaneous flame spread rate and the visible flame structure are compared for a downward spreading flame over ashless filter paper with the corresponding stationary flame. The results indicate that the difference between the two configurations are within experimental uncertainties and the stabilized flame can represent a spreading flame adequately, including variability of flame spread rate and the flame geometry, for further observation. B. Preparation for Space-Based Experimental Work: We have refined the objectives of our space-based experiments and produced a concept design for the proposed experiment. The hypothesis to be tested in the space-based experiments are the following: a. Flame spread rate over a thin fuel sample can be predicted by the closed form expression developed by our theoretical work in the past, including the effect of fuel thickness, pressure, and flow velocity. b. Under certain conditions (high fuel thickness, low ambient pressure, and low flow velocity) a steady flame spread cannot be sustained even at high oxygen level. c. The separation between the thermal and species fields is responsible for flame extinguishment and may provide clues to the time of extinction for unsteady flames. To achieve these goals we propose an experiment; the basic concept is as follows: A carousal driven by a stepper motor holds several sample holders, each of which are placed in the test zone by rotating the carousal. Samples of different thicknesses are held in each sample holder, which carries its own igniter. As a sample is rotated into the test zone, the igniter snaps into the ignition circuit. As the flame spreads over the sample, gas phase thermocouples and CO2 sensors monitor the propagation of the thermal diffusion fronts. Also, instantaneous propagation speed of the thermal front is recorded by a flame tracker mechanism like the one discussed in Sec. A.2. Once the propagating flame reaches the end of the sample or extinguishes, the recording of the temperature and species field will continue to provide valuable data on transient cooling of the gas phase and the diffusion of species. After a preset interval, the next sample is rotated into position and the experiment is repeated for a new fuel thickness. The carousel will then be replaced with a second carousel and this time a jet flow facility used by other researchers in their experiments will be utilized to created prescribed flow velocities. C. Theoretical/Modeling Work: We have made considerable progress in modeling flame spread over solid fuels under different conditions. Our modeling/theoretical effort can be summarized as follows: 1. We have been developing Web based tools for calculating equilibrium temperature of PMMA and Cellulose combustion. This calculation tools, which can be used by the community, helps us determine exactly how much sample burn is possible (under different conditions) in a closed chamber without significantly altering the oxygen level. 2. We are using a two dimensional model with finite-rate one-step kinetics, and radiation to simulate opposed flow flame spread. The model clearly shows the separation between the thermal and species field for flame close to extinction. It also supports the theoretical prediction about the existence of a critical fuel thickness beyond which steady spread is not sustainable. 3. By comparing narrow band calculations with total calculations, we have established that total radiation calculation is sufficient for accurate radiation modeling. Radiation signature, as measured by the percentage of heat generated that is lost by radiation, has been calculated showing the importance of radiation in the microgravity regime. 4. We are modeling downward flame spread in preparation for a comparison with the data acquired from the stabilizer apparatus as well as the flame tower. D. Dissemination of Results: In this period, we have presented seven conference papers and two journal papers have been submitted, with one accepted for publication. The equilibrium calculator has been posted on the web (http://www.thermofluids.net/) for the benefit of the community.
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